EDUVISIONAI : AI-POWERED HISTORY E-LEARNING SYSTEM FOR VISUALLY IMPAIRED O-LEVEL STUDENTS IN SRI LANKA

25-26J-438

Project Proposal Report

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Software Engineering

Department of Computer Science & Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

EDUVISIONAI : AI-POWERED HISTORY E-LEARNING SYSTEM FOR VISUALLY IMPAIRED O-LEVEL STUDENTS IN SRI LANKA

AUDIO-BASED LEARNING MODULE FOR O/L HISTORY EDUCATION – TTS WITH EMOTIONAL TONE & NLP SIMPLIFICATION

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: Date 2025/8/28

Signature of the co - supervisor: KR. Date 2025/08/28.

ABSTRACT

Visually impaired students face major barriers in history education because textbooks and classroom instruction rely heavily on visual material (maps, photos, timelines). This project develops an AI-powered audio storytelling module that transforms Sri Lankan O/L history texts into vivid, culturally contextualized audio lessons enriched with expressive text-to-speech, narrative restructuring, and carefully timed sound effects (ambient and event sounds). The system emphasizes accessibility (screen-reader integration, simple audio navigation, offline playback) and pedagogy (age-appropriate simplification, guided scaffolding, formative quizzes). We will design and train an NLP pipeline to convert textbook passages into descriptive, emotionally appropriate narratives; integrate high-quality TTS with prosody control; create an SFX library contextualized to Sri Lankan history; and evaluate the system's impact on engagement, comprehension, and retention with controlled user testing among visually impaired O-Level students.

Keywords: visually impaired education, history learning, natural language processing, expressive TTS, audio design, accessibility, Sri Lanka, O-Level.

LIST OF ABBREVIATIONS

O/L	Ordinary Level
TTL	Text-To-Speech
AI	Artificial Intelligence
NLP	Natural Language Processing
ML	Machine Learning
IDE	Integrated Development Environment
VCS	Version Control System
UAT	User Acceptance Testing

Table 1 - List of Abbreviations

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1. INTRODUCTION

1.1 Background

O/L History students, particularly those who are visually impaired, require accessible methods to grasp complex historical narratives and cultural contexts to understand key events and succeed in exams. This content forms the foundation for their historical knowledge, enabling them to appreciate Sri Lanka's rich heritage. However, the volume of events, dates, and descriptions can be overwhelming, especially without visual aids, making it challenging for visually impaired students to retain this information over time.

Without immersive and multisensory reinforcement, much of this learned history fades away, leading to a significant loss of knowledge shortly after exams. This is particularly problematic for visually impaired students who may pursue further studies or careers involving history or cultural fields, as the foundational narratives they once understood become less accessible, hindering progression to advanced topics.

The ability to retain information is greatly affected by factors such as the mode of delivery, sensory engagement, and the interval since last exposure, as shown in studies on multisensory learning. Multisensory approaches, which engage multiple senses, have been shown to enhance memory retention by creating richer cognitive associations [2]. One effective method is to incorporate audio narratives with culturally relevant sound effects to simulate historical experiences over time.

Traditionally, systems like audio books and braille texts have been used for accessible education [3]. These methods were effective in manual, non-digital eras. However, the rise of AI-powered platforms offers new possibilities. These platforms generate vast amounts of adaptive content, which can be leveraged to train sophisticated models for multisensory learning.

The AI-driven multisensory model is a technique designed to predict engagement and retention by delivering immersive audio content [4]. The model uses NLP to generate narratives and synchronizes them with sound effects, considering factors like user feedback and comprehension outcomes. Consequently, the model dynamically adjusts

content delivery to occur when the learner is most receptive, enhancing long-term retention.

Several enhancements can be made to the multisensory model to tailor it for Sri Lankan history education [4]. First, incorporating cultural semantics between historical events can improve the model. Sri Lankan history often involves interconnected eras, so experiencing one event aurally can reinforce others. For instance, hearing temple bells for an Anuradhapura-era narrative might aid recall of related concepts like ancient architecture and royal reigns. By mapping these correlations, the model can adjust narratives to reinforce clusters of related events, boosting learning efficiency.

Another key enhancement is adapting the model to varying complexity levels of historical concepts. Sri Lankan history spans simple timelines to intricate socio-political events requiring different cognitive efforts. The model can be tuned to provide richer sound effects for complex topics. While straightforward events might need basic audio, complex ones like battles could benefit from layered soundscapes. The model can also manage sensory load by distributing elements, preventing overload and improving assimilation.

1.2 Research Gap

Multisensory educational systems are increasingly used, but there remains a significant gap in tailoring these for visually impaired students in specialized subjects like history, particularly with audio immersion [1]. While effective for general accessibility, traditional tools like audio descriptions and braille often lack the immersion needed for the cognitive demands of historical storytelling. In history, where events are built on cultural and temporal links, these systems typically deliver content linearly without accounting for sensory enhancements. This limitation highlights the need for advanced models that dynamically adapt to the unique attributes of historical content.

Moreover, despite advancements in AI for education, little research has explored optimizing these models for domain-specific applications like Sri Lankan history for the visually impaired [5]. Current systems often overlook the challenges posed by cultural contexts. For example, historical events in Sri Lanka are linked by themes of heritage and conflict, suggesting that audio enhancement for one event could aid others. However, most models fail to exploit these semantic links, missing opportunities to enhance immersion through synchronized multisensory elements.

Another critical gap is personalizing multisensory delivery based on learner profiles, especially for visually impaired users in demanding subjects like history [4]. While some progress exists in adapting content to performance, limited studies address factors such as prior knowledge, event complexity, and sensory preferences. For instance, abstract events might require richer audio cues than familiar ones. Developing a model incorporating these could make multisensory education more effective in specialized areas.

Large-scale AI platforms produce abundant data for training complex models, yet in the context of audio-enhanced history for visually impaired O/L students in Sri Lanka, this potential is largely untapped [7]. Existing models focus on broad accessibility patterns without delving into how cultural nuances affect engagement. There is a clear need for research using domain-specific data to create targeted multisensory techniques, potentially revolutionizing accessible history education [5][6].

Platform	Focus on Sri Lankan O/L History	Multi-Sensory Storytelling(Audio + Sound Effects)	Accessibility for Visually Impaired
Audemy	**NO**	**NO**	**YES**
REMathEx	**NO**	**NO**	**NO**
Animal Watch	**NO**	**NO**	**YES**
EduVisionAI	**YES**	**YES**	**YES**

Table 2 - Competitive Analysis

1.3 Research Problem

Despite advancements in AI and multisensory tools, current educational systems exhibit notable limitations in supporting immersive learning of cultural history for visually impaired students, such as in Sri Lankan O/L curricula. Traditional approaches, like audio books and basic texts, often use static delivery methods and fail to adequately incorporate semantic links or audio enhancements inherent in historical narratives. Furthermore, insufficient research has examined tailoring these systems to diverse sensory needs and varying difficulty levels of events.

The study aims to develop and evaluate an enhanced multisensory storytelling module that integrates domain-specific factors like cultural links between historical events and learner traits such as sensory preferences and cognitive load. This module should utilize AI and NLP to create adaptive narratives with sound effects, fostering a personalized schedule that improves retention and comprehension. The objective is to determine how these domain-specific and learner-centric features can be embedded into multisensory systems to elevate learning outcomes in accessible history education.

2. OBJECTIVES

2.1 Main Objective

To develop an AI-powered e-learning system with a multisensory storytelling module that uses descriptive audio narratives to deliver engaging Sri Lankan history lessons for visually impaired O/L students

2.2 Specific Objectives

1. Design and train an NLP-based AI model to generate vivid, descriptive audio narratives of historical events using Sri Lankan history texts.
2. Create culturally relevant audio scripts with sound effects (e.g., temple bells, battle sounds) to enhance immersion.
3. Evaluate the module's effectiveness through user testing with visually impaired students to measure engagement and comprehension.

3. METHODOLOGY

3.1 Requirement Gathering

The requirement gathering process involved a comprehensive review of existing research, analysis of current systems, and examination of practical scenarios related to multisensory learning and accessible history education.

3.1.1 Past Research Analysis

A thorough examination of prior research was conducted to understand the evolution and limitations of multisensory educational tools, particularly for visually impaired learners. This analysis focused on methods and technologies used in previous studies, especially those involving AI integration and audio enhancements. Key interests included the effectiveness of audio combinations and recent data-driven innovations. The review pinpointed gaps, such as the need for models incorporating cultural semantic links in history for the visually impaired.

3.1.2 Identification of Existing Systems

An assessment of existing accessible e-learning platforms was performed to identify strengths and weaknesses. This included surveying popular tools for visually impaired education and evaluating their applicability to history learning. The analysis aimed to determine how these systems manage sensory content and potential improvements. Insights from this informed the design of the proposed solution.

3.2 Feasibility Study

3.2.1 Technical Feasibility

3.2.1.1 Knowledge on Technologies

To develop an AI-driven multisensory storytelling module, understanding several technologies is essential:

1. **AI and NLP Models**: Expertise in natural language processing for generating narratives, using models like GPT or BERT.
2. **Audio Processing**: Proficiency in tools for creating and editing sound effects and narratives.
3. **Adaptive Algorithms**: Understanding algorithms for personalizing content based on user feedback and comprehension.

3.2.1.2 Knowledge on Tools

Developers should be proficient in using the following tools when developing the component,

- **Programming Languages**: Python for developing algorithms and models, JavaScript for programming.
- **Frameworks**: React framework for frontend development, Flask for backend development, Tailwind libraries and utilities for UI.
- **AI Libraries**: Hugging Face Transformers, TensorFlow for NLP.
- **Audio Tools**: Audacity or similar for sound design.
- **Project Management**: Jira for tracking progress and managing tasks

3.2.1.3 Data Collection Knowledge

Effective data collection and preprocessing are crucial for the success of the project:

1. **Data Collection**: Sourcing Sri Lankan history texts, Grade 10 & 11 textbooks, and accessibility assessments.
2. **Data Preprocessing**: Cleaning texts, annotating for NLP, and preparing audio mappings.

3.2.2 Schedule Feasibility

The proposed model will be developed on a systematic timescale, as indicated in the project plan. This calendar will ensure timely progress and adherence to project milestones by establishing defined deadlines and deliverables.

3.2.3 Economical Feasibility

The project's goal is to create a cost-effective solution by utilizing current technology and reducing expenditures. A precise budget will be created to account for development, testing, and deployment expenses, taking into account future economic fluctuations.

3.3 System Analysis

3.3.1 Software Solution Approach

The development of the proposed spaced repetition model involves the following steps:

1. **Data Collection**: Gather history texts, learner performance and vocabulary difficulty through surveys and assessments.
2. **Model Design**: Create NLP model for narratives, integrate audio effects.
3. **Feature Extraction**: Identify cultural elements for audio mapping.
4. **Model Training**: Train on texts to generate immersive content.
5. **Model Evaluation**: Test with users for refinement.

3.3.2 Tools & Technology

- **Programming Languages**: Python, JavaScript
- **AI Frameworks**: TensorFlow, Hugging Face
- **Data Tools**: Pandas, NumPy
- **Web Frameworks**: React JS
- **UI Design**: Figma
- **Diagramming**: Draw.io
- **IDE**: VS Code
- **VCS**: Git (GitHub)
- **Collaboration**: Microsoft Teams
- **Testing**: Jest, Postman
- **Deployment**: Docker, Kubernetes, Jenkins
- **Project Management**: JIRA for tracking progress and managing tasks

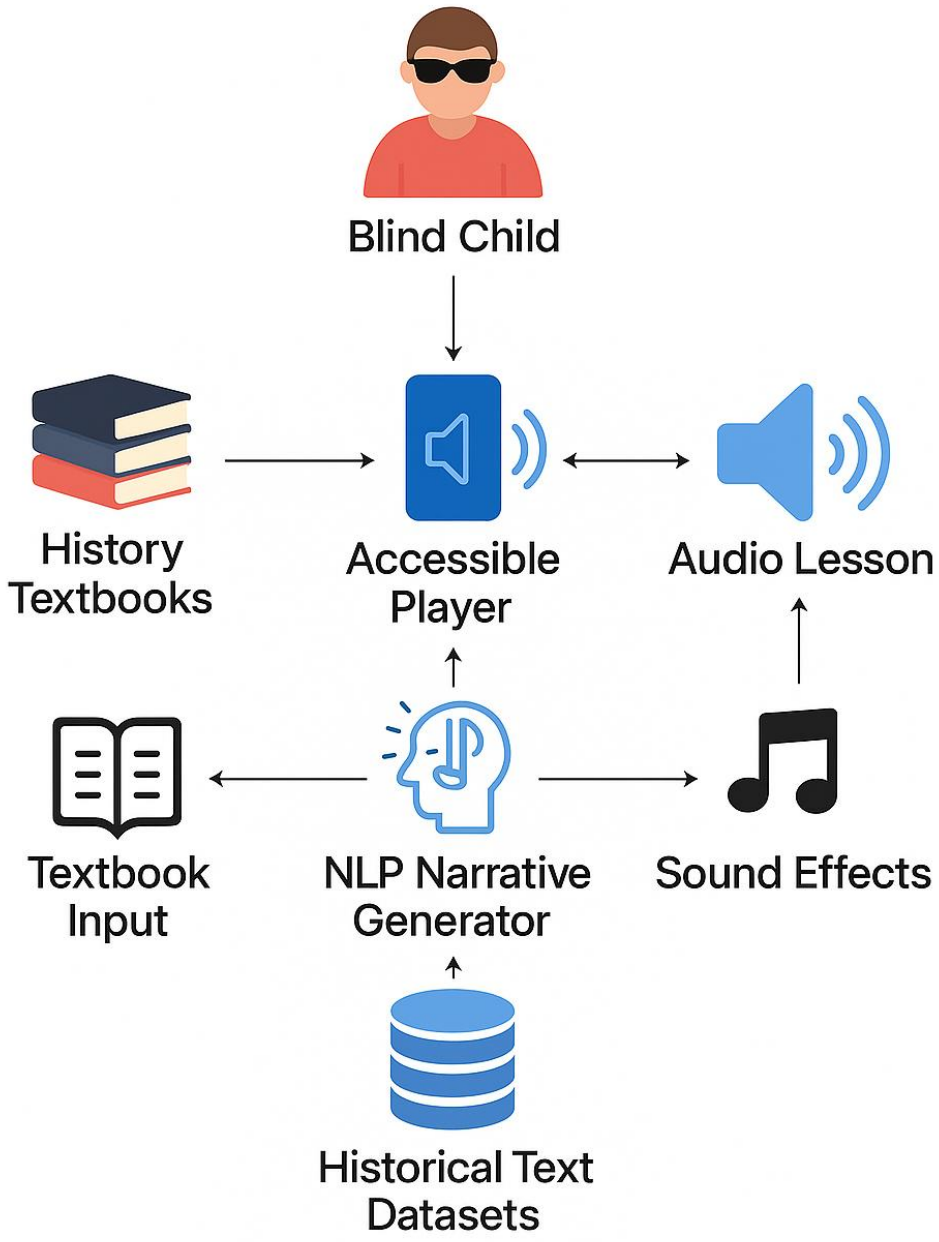

Figure 1 - System Diagram

3.4 Project Requirements

3.4.1 Functional Requirements

1. **User Registration and Authentication** a. The system must allow students to register using email or accessible methods. b. Users must log in securely to access personalized lessons.

2. **Multisensory Storytelling Module** a. The application must provide AI-generated audio narratives with sound effects for history lessons. b. Users must experience adaptive content based on progress. c. The system must track interactions to optimize delivery.

3. **Performance Tracking and Feedback** a. The application must monitor progress, including comprehension and engagement. b. Users must receive audio feedback on performance.

4. **Adaptive AI Algorithms** a. The system must use NLP to generate and adjust narratives. b. The algorithm must incorporate cultural elements in sound design.

5. **Content Management** a. Administrators must add/update history content. b. The system must support topic-based lesson sets.

6. **Audio Features** a. The application must include descriptive narratives with sound effects. b. Provide immediate feedback on user interactions.

7. **User Interface** a. The interface must be accessible, voice navigable. b. Compatible with screen readers.

8. **Data Export and Reporting** a. Users must export progress data. b. Generate audio reports on learning.

3.4.2 Non-Functional Requirements

1. **Performance and Scalability**: Handle multiple users without lag,

   Scale for growing content and users.

2. **Security:** Secure storage of user data.

   Encryption for sensitive information.

   Comply with data protection.

3. **Usability:** Intuitive for visually impaired users.

   Include voice commands.

4. **Reliability and Availability**: Minimal downtime.

   Backup and recovery plans.

5. **Maintainability:** Modular design for updates.

   Clear documentation.

6. **Localization:** Support Sinhala/Tamil if needed, focused on English medium.

   Adapt to cultural contexts.

7. **Accessibility**: Follow WCAG standards.

   Features like audio descriptions.

3.5 Testing

Testing is a vital part of developing a reliable software system, ensuring it functions as intended and meets all specified requirements. This process involves several stages to identify and resolve issues that could affect performance, usability, and reliability. Unit testing starts by examining individual components, such as the NLP module, to confirm they work correctly in isolation. Next, integration testing checks that these components, like audio sync and NLP, work seamlessly together, addressing issues in data flow and communication. System testing then evaluates the entire system to verify it meets both functional and non-functional requirements, including performance, security, and compatibility. Finally, user acceptance testing (UAT) involves visually impaired users to validate accessibility and engagement, ensuring the system meets real-world needs. Through these steps, testing delivers a robust, user-friendly system ready for deployment, free of defects, and capable of providing a seamless experience.

3.6 Timeline

Figure 2 - Project Grantt Chart

3.7 Risk Management Plan

3.7.1 Introduction

Risk Management is vital for project success. This plan identifies risks and mitigation for team, panel, and supervisor interactions.

3.7.2 Risk Identification

Table 3 - Risk Identification

Risk Category	Risk Description	Mitigation Strategies
Team-Related Risks		
Skill Gaps and Knowledge Deficiencies	Lack of necessary skills or knowledge withing the team	Early identification of skill gaps, training sessions, workshops, external experts, mentors.
Team Member Availability	Unexpected unavailability of team members due to illness, emergencies, or commitments.	Cross training, contingency plans, task reallocation.
Communication Breakdown	Miscommunication or lack of communication among team members.	Clear communication channels, team meetings, project management tools, documentation.
Conflict Among Team Members	Disagreements or conflicts within the team.	Conflict resolution mechanisms, feedback sessions, mediation, open communication, mutual respect.
Project Scope Creep	Project expansion beyond its original scope.	Adherence to project charter, formal change management process.
Panel/Supervisor-Related Risks		
Misalignment of Expectations	Mismatch between project team's output and panel/supervisor's expectations.	Regular meetings, feedback, alignment with expectations.
Delayed Feedback or Decision Making	Slow or delayed feedback from the panel or supervisor.	Clear deadlines, reminders, proactive feedback seeking.

Changes in Panel/Supervisor's Availability	Unavailability of panel or supervisor due to other commitments.	Advance scheduling, alternative contacts, contingency plans.
Scope or Requirement Changes from Panel/Supervisor	Changes to project requirements or scopes introduced by the panel or supervisor.	Formal change control process, assessment of impacts, negotiation.

3.7.3 Risk Monitoring and Control

Risk monitoring and control is an essential process to ensure the system's stability and success throughout its lifecycle. This involves continuously identifying, assessing, and prioritizing potential risks, such as technical failures or user accessibility issues, to minimize their impact. Regular tracking of risk indicators, like performance metrics or user feedback, allows for timely detection of emerging threats. Mitigation strategies, such as adjusting system configurations or enhancing support for visually impaired users, are implemented as needed. By maintaining a proactive approach, risk monitoring and control ensures the system remains reliable, secure, and aligned with user needs, adapting to challenges as they arise.

3.7.4 Conclusion

This project's success depends on effective risk management. By identifying possible risks early on and developing clear methods for managing them, the project team may limit the possibility of interruptions and ensure that the project is finished on time, within scope, and to the satisfaction of the panel and supervisor(s).

3.8 Communication Management Plan

3.8.1 Communication Mediums

- **Email:** Used for formal updates, sharing documentation, progress reports, and detailed communications that require record-keeping.
- **WhatsApp:** Used for informal, quick exchanges, real-time updates, urgent matters, and brief discussions between team members.
- **Meetings:** Scheduled weekly or as needed for in-depth discussions, problem-solving, and decision-making regarding project development.

3. WORK BREAKDOWN

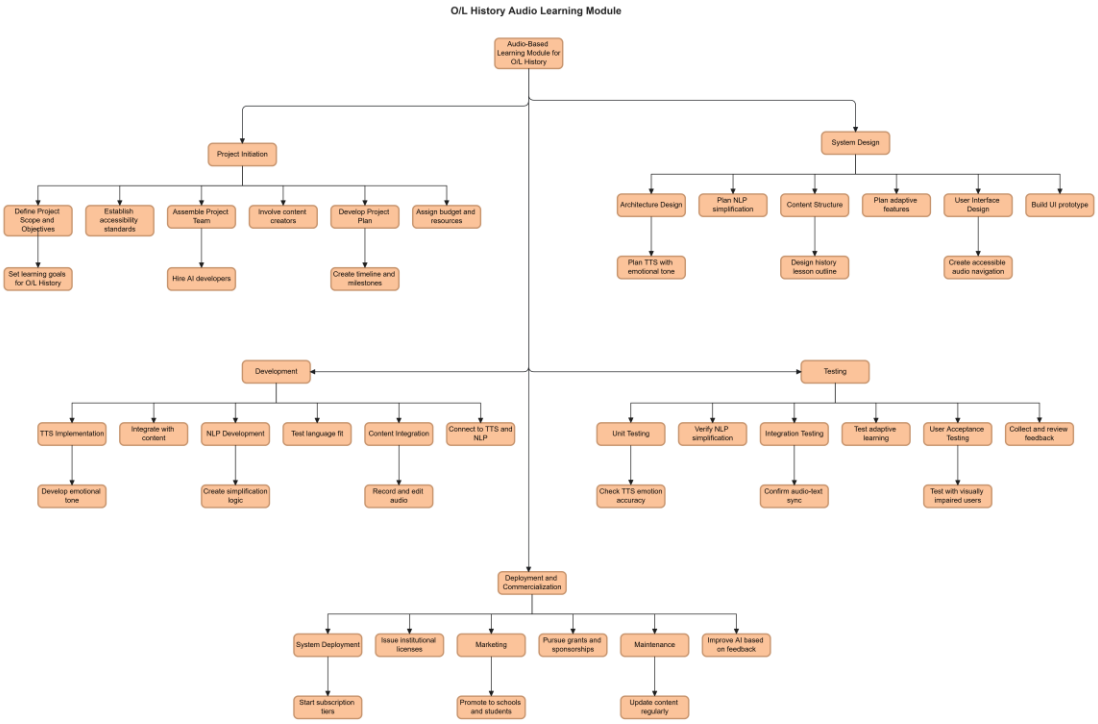

Figure 3 - Work breakdown structure

5. COMMERCIALIZATION

The commercialization of the audio-based learning module for O/L History education, featuring text-to-speech (TTS) with emotional tone and NLP simplification, aims to generate sustainable revenue while ensuring accessibility for visually impaired learners.

1. Subscription-based model includes tiers for schools, institutions, or individual students:
    - **Basic Tier:** Free access to core TTS features with limited history content.
    - **Premium Tier:** Full access to emotionally toned TTS, advanced NLP simplification, and expanded history lessons
2. Institutional licensing offered to educational boards or schools for bulk access, promoting widespread use at a reduced cost per user.
3. Grants and sponsorships sought from organizations supporting inclusive education or disability initiatives to subsidize free access for visually impaired learners unable to pay.
4. Non-intrusive, educationally relevant advertising may be considered, such as promotions for accessible historical resources, to enhance revenue without disrupting the learning experience.

6. BUDGET

Types	Cost
Internet use and web hosting	LKR 20,000.00
Training Cost	LKR 15,000.00
Publication Cost	LKR 50,000.00
Other	LKR 10,000.00
Total	LKR 95,000.00

Table 4 - Cost management plan

The cost can vary depending on several factors when implementing the final product.

7. Summary

The audio-based learning module for O/L History education, incorporating text-to-speech (TTS) with emotional tone and NLP simplification, is designed to enhance accessible learning for visually impaired students in Sri Lanka. The project initiates with defining scope, assembling a team of AI developers and content creators, and planning resources. System design focuses on creating TTS and NLP architectures, structuring history lessons with adaptive features, and designing an accessible audio interface. Development includes implementing emotional TTS, building NLP simplification, and integrating audio content. Testing ensures accuracy through unit, integration, and user acceptance phases with visually impaired users. Deployment and commercialization involve launching subscription tiers and institutional licenses, marketing to schools, and maintaining content and AI enhancements. The module promises a sustainable, high-quality learning experience.

References

[1] Maćkowski, M. S., Brzoza, P. F., & Spinczyk, D. R. (2018). Tutoring math platform accessible for visually impaired people. *Computers in biology and medicine*, *95*, 298–306. https://doi.org/10.1016/j.compbiomed.2017.06.003

[2] Tóth, V., & Parkkonen, L. (2019). *Autoencoding Sensory Substitution*. arXiv. https://arxiv.org/abs/1907.06286

[3] Jackson, R. M. (2021). *Audio-Supported Reading for Students Who Are Blind or Visually Impaired*. Wakefield, MA: National Center on Accessible Educational Materials (CAST) https://aem.cast.org/binaries/content/assets/common/publications/aem/asr-blind-visuallyimpaired.pdf

[4] Yang, C., & Taele, P. (2025). *AI for Accessible Education: Personalized Audio-Based Learning for Blind Students*. In *CHI 2025 Workshop on Augmented Educators and AI*, Yokohama, Japan. https://arxiv.org/abs/2504.17117

[4] Sarwar, S., Turab, M., Channa, D., Chandio, A., Sohu, M. U., & Kumar, V. (2022). *Advanced Audio Aid for Blind People*. arXiv. https://arxiv.org/abs/2212.00004

[5] A. M. Senavirathna, C. Premachandra, and S. Wijesinghe, "Low-cost Talking Calculator for Visually Impaired People in Sri Lanka," in *Proc. Int. Conf. Image Process. Robot. (ICIPRob)*, Negombo, Sri Lanka, 2020.

[6] M. A. T. Najaah and A. S. Thowfeek, "Voice-based online examination system for visually impaired," *Journal of Management*, vol. 16, no. 1, pp. 27–38, Apr. 2021, ISSN: 1391-8230

[7] Maćkowski, M., & Brzoza, P. (2022). Accessible Tutoring Platform Using Audio-Tactile Graphics Adapted for Visually Impaired People. *Sensors (Basel, Switzerland)*, *22*(22), 8753. https://doi.org/10.3390/s22228753